Concept Table - Social Loafing
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Table 1

Table of core constructs

ID	Type - Phenomenon - Concept - HOC	Short name	Quote	Reference	Rel. type (n, p, i, r, ...)	Comment (e.g., identified gaps & inconsistencies)	Incl. (Y/N)?
CT	C	Co-Targets	-"The more people who are the target of this pressure, [...]" (S.823) - "According to Latane's (1981) social impact theory, people can be viewed as […] targets of social impact." (S. 682) AND - "and number of group members" (Latané et al., 1979, S. 823) - "impact is divided among the group members" (Latané et al., 1979, S. 830) - "However, if a person is a member of a group that is the target of social forces from outside the group" (Latané et al., 1979, S. 823)	Latané et al., S.823 (1979) AND Karau & Williams, S. 682 (1993)	n1, p1,p2, n12		Y
S	C	Source of pressure	- "The impact that the experimenters have on an individual [...]" (S.825) - "According to Latane's (1981)	Latané et al., S.825 (1979) AND	n2, p2	these persons are not described as a pressure precisely,	Y

			social impact theory, people can be viewed as either sources […] of social impact.” (S. 682)	Karau & Williams , S. 682 (1993)		but we can assume that they are the reason why the pressure occurs we set: pressure = impact	
CON	C	Contingency between input and outcome	“[...] the contingency between their input and the outcome [...]” (S. 830)	Latané et al., S. 830 (1979)	p1, n3, t1	here, contingency is simply explained through the word itself, no further definition	Y
INC	C	Incentive	“[...] incentive to work equally hard in groups of different sizes.” (S.830)	Latané et al., S. 830 (1979)	t1, n4, p3		Y
P	C	impact/ pressure	“the amount of social pressure on the target person” (S. 823) “pressures on each individual” (S. 830) / “In a group performance situation in which pressures to work come from outside the group” (S. 830) AND “[...]the impact of these forces on any given member [...]” (S.823)	Latané et al., S.823 (1979)	p2, n5, n6, p3	impact and pressure are no really distinctive concepts, cause impact can officiate as a social facilitation. here, we assume its the same cause we limited the	Y

						scope to the negative form of social impact, no facilitatio n at all	
IE	C	individual effort	“individual effort” (S. 822 ff.)	Latané et al., S.822 ff. (1979)	p3, n7, t2	In this paper, there is no real distinction between “effort” and “performa nce”. These concepts were used partially synonymo usly. The problem is, they were not defined.	Y
Psy	HO C	Psyche	no citation	/	n11	We defined this HOC as the space in which the mental process is defined	Y
IO	C	individual output	“in this case n times the average individual output” (S. 825) AND “Since individual scores” (S. 830) AND - “leading to an apparent drop in	Latané et al., S.825/S. 829/S.83 0 (1979)	n13, n14, n15, t2		

			individual performance, a phenomenon we call social loafing" (S. 825) - "sum of their individual performances" (S. 829)				
MC	C	maximum capacity	"dyads pulled at 91% of the sum of their individual **capacities**, trios at 82%, and groups of six at only 78%" (S. 825)	Latané et al., S.825 (1979)	n16, t2	maximal capacity of individual, when 100 sources and 0 cotargets are present (no citations are available, this is our own interpretation to simulate the phenomenon)	Y
ME	HOC	Matching of effort	"this effort matching interpretation" (S.1204) "they will maintain equity by matching the effort levels they expect from their partners" (S.1200) "social loafing occurs because individ uals expect others to slack off in groups and, therefore, reduce their own efforts to maintain equity" (S.683)	Harkins & Jackson, S.1200/1204 (1985) AND Karau & Williams, S. 683 (1993)	/	this effect occurs only when the scope is limited to holding identifiability constant throughout the experiment	N
EMC	C	Effort condition	"In the HE condition, she said she had tried very hard on the practice trial and, because she thought the research was interesting, she was	Harkins & Jackson, S.1202	n17, t3	this is purely a condition construct,	Y

			going to try hard throughout the experiment; in the LE condition, she said she had not tried on the practice shout and, because she thought the research was boring, she wasn't going to try hard during the rest of the experiment; and in the SLR condition, she simply made small talk until the experimenter returned" (S.1202)	(1985)		no explicit values were give in Harkins & Jackson (1985), manifest variable outside the Psy-Box	
LE	C	loafing expectation	"expectations of the amount of partner effort" (S.1204) AND "expectations concerning the partner's intended levels of effort"	Harkings & Jackson, S.1204 (1985)	n18, t3, p3	no values were given here either, but this is strictly a latent variable - we transform it from the effort matching condition to have a construct that matches the condition in the Psy-Box	Y
SL	P	Social loafing	"people exhibit a sizeable decrease in individual effort when performing in groups as compared to when they perform alone"	Latané et al., S.822 (1979)	i1	this is the core phenomenon for the report	

Table 2

Table of variables.

Construct	Source (if given)	Scale level	Range/Values	Anchors
CT	"and *number* of group members" (Latané et al., 1979, S. 823)	discrete	$\in \mathbb{N}_0$	0 = "zero cotargets are present"; n = "number of cotargets (other persons in the evaluated group, number of group members minus 1)"
S	"*number* of these other persons." (Latané et al., 1979, S. 823) AND "Thus, as the *number of performers decreased, the size of the audience increased*. According to Cottrell's evaluation apprehension hypothesis (1972), the presence of an evaluative audience should enhance performance for a simple, well learned task such as rope pulling, and, although there is little supportive evidence, it	discrete	$\in \mathbb{N}_0$	"0 = "zero persons that cause the social pressure"; n = "number of persons that cause the pressure"

	seems reasonable that the *larger the audience*, the greater the enhancement“ (Latané et al., 1979, S. 826)			
P		continuous	∈ [0; 1[	0 = “zero persons are source of social pressure”, “no social pressure” 1 = “maximal social pressure (a theoretical amount on infinite spectators are present)”
CON	“Individuals could "hide in the crowd" (Davis, 1969) and avoid the negative consequences of slacking off, or they may have felt "lost in the crowd" and unable to obtain their fair share of the positive consequences for working hard.” (Latané et. al, 1979, S. 830) “reward” (Latané et. al, 1979, S. 830)	continuous	∈ [0; 1]	0 = “expectation that the share (reward/punishment) of each member is completely independent of what they’ve given” 1 = “expectation that their share (reward/punishment) is equal to what they’ve given”
INC		continuous	∈ [0; 1]	0 = “no incentive to give any individual effort” ; 1 = “incentive to give full effort”

IE	e. g. “dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%” (Latané, 1979, S. 825)	continuous	∈ [0; 1]	0 = “the person shows no effort at all to bring in some contribution to the group task”, 1 = “the person shows their maximal effort in contributing to the group task”
O	“sum of their individual capacity” (Latané, 1979, S. 824); e. g. “dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%” (Latané, 1979, S. 825)	continuous	∈ [0; 1]	0 = “maximal social loafing effect, nobody does anything”; 1 = “sum of the maximal individual performances” “no social loafing effect” “everybody gives their best” o = “percentage of the theoretical maximal outcome (sum of each individual capacity) of the whole group”
IO	“dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%” (Latané, 1979, S. 825)	continuous	∈ [0; inf[	0 = dynes/cm^2 n = dynes/cm^2
MC		continuous	$\in \mathbb{R}_0$	depends on task (clapping with hands, rope pulling)
EMC	“she said she had tried very hard	ordinal	1, 2 or 3 is possible	1 = low effort condition,

	on the practice trial and, because she thought the research was interesting, she was going to try hard throughout the experiment" (Jackson & Harkins, 1985, S.1202) AND "she said she had not tried on the practice shout and, because she thought the research was boring, she wasn't going to try hard during the rest of the experiment" (Jackson & Harkins, 1985, S.1202) AND "she simply made small talk until the experimenter returned" (Jackson & Harkins, 1985, S.1202)			2 = Social loafing replication condition, 3 = high effort condition
LE	view "EMC" for citations, they're identical	ordinal	"match_low", "no_info", "match_high"	"match_low" = the subject thinks the Co-Targets were giving their full effort in the practice trial, "no_info" = the subjects has no information whatsoever about the others goal in effort,

“match_high” = the subject thinks the Co-Targets had not tried hard in the practice trial

Table 3

Table of relationships.

Output variable	Function to compute the output variable	Comment	To what extent derived from theory?[a]
relationship P1 => CON	CON = 1/(CT+1) => CON = 1/n	quote q3 mentions the "reward" but if each group member performed up to its own capacity would feed an inconsistency between the effort (100%) and the reward/outcome (1/n)	CT → CON: q1: "We believe that social loafing occurs because there is usually a stronger perceived contingency between individual effort and valued outcomes when working individually. When working collectively, factors other than the individual's effort frequently determine performance, and valued outcomes are often **divided among all of the group members**." Karau & Williams, S. 684 (1993) q2: "which both suggest that individuals' outcomes are more contingent on their efforts when working with others who are expected to perform poorly." Karau & Williams, S. 696 (1993) q3: "the reward is psychologically divided equally among the participants, each getting 1/n units of reward" (Latané, 1979, S.830)
relationship P2 => P	P = S/(S+CT+1)		q4:"[...]the amount of social pressure on the target person should increase as a multiplicative function of the strength, immediacy, and number of these other persons." (Latané, 1979, S.823)

			=> derived from theory q5: “Impact is divided up among the group members, in much the same way that responsibility for helping seems to be divided among witnesses to an emergency” (Latané, 1979, S.823) => derived from theory q6: “If a person is the target of social forces, increasing the number of other persons also in the target group should diminish the pressures on each individual because the impact is divided among the group members.” (Latané, 1979, S. 830) CT → P: q7: “The more people who are the target of this pressure, the less pressure should be felt by any one person.” (Latané, 1979, S. 823) => derived from theory q8: “increased group size to result in reduced effort” (Latané, 1979, S. 823)

relationship P3 => IE	get_IE <- function(INC, P, LE) {
	  # the Social Loafing baseline (Latané)
	  # whilst CT increases, the output decreases
	  loafing_effort <- 0.5 * INC + 0.5 * P
	
	  # the effort matching logic
	  # we're definign fix values which represent the expectation for the "alone"-
	  # condition of other
		we added the effort matching condition/variable later on, no real mathematical relation could be found in the underlying paper
		
		this function includes one of two sources of variability between the subjects cause
			q9: “Latane further suggests that just as psychophysical reactions to external stimuli can be described in terms of a power law (Stevens, 1957), so also should reactions to social stimuli, but with an exponent having an absolute value less than 1, so that the nth person should have less effect than the (n - l)th.” (Latané, 1979, S. 823)

subjects (independent of the current group size)

```
  final_IE <- case_when(
    # condition: co-Target is not loafing in its "alone"-condition
    LE == "match_high" ~ 0.9,

    # condition: Co-Target is loafing in its "alone"-condition
    LE == "match_low"  ~ 0.4,

    # In der no_info Bedingung greift das klassische Modell (Loafing)
    LE == "no_info"    ~ loafing_effort,

    TRUE               ~ loafing_effort
  )

  return(final_IE)
}
```

we added the noise term on top of it (but only in the superfunction version of it)

For this function, it was reasonable to present the full R code.

=> derived from theory

P → IE:
q10: “The more people who are the target of this pressure, the less pressure should be felt by any one person. Since people are likely to work hard in proportion to the pressure they feel to do so, we should expect increased group size to result in reduced efforts on the part of individual group members.” (Latané, 1979, S. 823)

q11: “If a person is the target of social forces, increasing the number of other persons also in the target group should diminish the pressures on each individual because the impact is divided among the group members. In a group performance situation in which pressures to work come from outside the group and individual outputs are not identifiable, this division of impact should lead each individual to work less hard. Thus, whether the subject is dividing up the amount of work he thinks should be performed or whether he is dividing up the amount of reward he expects to earn with his work, he should work less hard in groups.” (Latané, 1979, S. 830)

q12: “in fact, participants matched their own efforts to these new expectations” (Jackson & Harkins, 1985,

			S.1199)
			q13: “by matching the effort levels they expect from their partners” (Jackson & Harkins, 1985, S.1200) => derived from theory
relationship t1 => INC	INC = CON		q14: “Thus, under these assumptions, if his own performance cannot be individually monitored, an individual's incentive to perform should be proportional to l/n.” (Latané, 1979, S. 830) => derived from theory
relationship t2 => IO	IO = MC * IE	since we need so compute a manifest outcome for each person and therefore for the experimental group means we had to multiply the percentage of individual effort with the maximum capacity	q15: “as with rope pulling, output can be measured in simple physical units that make up a ration scale” => derived from theory

relationship t3
=> LE
	get_LE <- function(EMC) {
	LE <- case_match(EMC,
	1 ~ "match_low",
	2 ~ "no_info",
	3 ~ "match_high"
	)
	return(LE)
	}
		we transformed the manifest construct/variable EMC into the latent construct/variable LE what created the need for such transformation	=> Not based on focal theory, but rather on common sense or other theories

[a] *Response scale:*

- *Dictated by theory (add IDs from the Construct Source Table as reference)*
- *Derived from theory*
- *Loosely inspired by theory*
- *Not based on focal theory, but rather on common sense or other theories*
- *Not at all*